ISABELA STATE UNIVERSITY • CITY OF ILAGAN CAMPUS
COLLEGE OF ENGINEERING, ARCHITECTURE, AND TECHNOLOGY | ICT DEPARTMENT

SYSTEM ALPHA AND BETA TESTING
MODULE–FUNCTION TEST MATRIX

List every system module and each individual function under that module. Use one row per function per test cycle.

A. SYSTEM INFORMATION

Information	Details	Information	Details
System / Project Title	Study Vault — Native Android Mobile Application (Java / XML)	**Version / Build**	v2.0.0 (Android SDK 24–35 / Gradle Kotlin DSL)
Development Team	_____ *(Leave blank)*	**Testing Date**	_____ *(Leave blank)*
Testing Cycle	[X] Alpha [X] Beta [X] Retest	**Environment**	[X] Local [ ] Staging [ ] Hosted
Tester / User Group	IT Students & Eligible IT Alumni (ISU-Ilagan)	**Device / Browser**	Android Emulator (API 35 / 10.0.2.2) & Physical Android Phone (Wi-Fi LAN)

B. MODULE INVENTORY

Before testing, list all modules in the system. Examples may include Login, User Management, Dashboard, Records Management, Reports, Settings, and Logout. Replace examples with the actual modules in your project.

Module ID	Module Name	Purpose / Main Feature	Functions Included (brief list)	Priority
MOD-01	**Splash & Session Routing**	Displays ISU/Study Vault branding and validates existing session token with server	Display splash branding, check EncryptedSharedPreferences token, validate profile status via /api/profile.php, route to Login or Main	High
MOD-02	**Native Login & Server Config**	Signs in approved IT students & alumni and configures XAMPP server URL	Validate Student ID & password, toggle password visibility, parse HTTP 4xx status banners, save Bearer token, configure Emulator/LAN URL	Critical
MOD-03	**3-Step Student ID Verification**	Verifies Student ID via 6-digit school email OTP and sets account password	Step 1 send_code, Step 2 verify 6-digit OTP & 60s resend countdown, restart with different ID, Step 3 preview verified record & set password	Critical
MOD-04	**Home Dashboard Fragment**	Greets authenticated student, shows plan status, stats, categories, and recent works	Display greeting & role badge, show active plan banner, hero search redirect, horizontal subject tiles, pull-to-refresh & retry state	High
MOD-05	**Research Archive & Search Fragment**	Searches and filters approved research with dynamic spinners and pagination	Keyword search with debounce, live autocomplete dropdown, dynamic Subject & Year spinners, Sort dialog, Clear Filters, Next/Prev page	Critical
MOD-06	**Research Details Activity**	Displays full thesis metadata, open abstract, clickable keywords, and access box	Load detail via /api/research-detail.php, display abstract to all users, keyword chip search, related study navigation, lock/unlock manuscript	Critical
MOD-07	**Access Plans & Demo Checkout**	Displays backend plans and executes prototype subscription activation	Load plans via /api/plans.php, show active/expired banner, launch CheckoutActivity, enforce demo checkbox, display receipt reference code	High
MOD-08	**Protected Native PDF Viewer**	Renders watermarked PDF pages natively using PdfRenderer and FLAG_SECURE	Enforce FLAG_SECURE screenshot lock, verify access via /api/manuscript.php, stream to private cache, stamp bitmap + overlay watermark, zoom/page	Critical
MOD-09	**Profile Fragment & Logout**	Displays verified ISU student details, subscription history, and logout dialog	Display student identity & verification status, show active plan card, list subscription history, open Server Config dialog, confirm & revoke token	High

C. FUNCTION-LEVEL TESTING MATRIX (USE ONE ROW PER FUNCTION)

Copy this table or continue on additional pages until every function in every module has been tested. For Alpha testing, the developer/team or internal tester verifies expected behavior. For Beta testing, intended users execute the same functions in realistic use and provide feedback.

Test ID	Module	Function / Action	Precondition / Test Data	Test Steps	Expected Result	Actual Result / Evidence	Status	Issue ID
TC-M01	**MOD-01 Splash**	Validate stored session token on app startup	Valid Bearer token in EncryptedSharedPreferences	1. Launch app 2. Observe SplashActivity	Calls /api/profile.php; routes to MainActivity if approved	Session verified; routed to HomeFragment	Pass	—
TC-M02	**MOD-02 Login**	Sign in with valid Student ID & password	Student ID: 24-01048, valid password	1. Enter ID & password 2. Tap Sign In	Receives Bearer token + Student JSON; opens MainActivity	Token saved in SessionManager; Home loaded	Pass	D-01
TC-M03	**MOD-02 Login**	Display server error banner for pending/invalid login	Pending account or wrong password	1. Enter credentials 2. Tap Sign In	Parses HTTP 401/403 JSON errorBody into tvAlertBanner	Exact server message shown in amber/red banner	Pass	D-01
TC-M04	**MOD-02 Login**	Configure XAMPP Server URL (Emulator vs. LAN IP)	Tap Server Connection Settings	1. Open ServerConfigDialog 2. Set 10.0.2.2 or LAN IP	Normalizes URL & rebuilds Retrofit client immediately	Base URL updated and persisted	Pass	—
TC-M05	**MOD-03 Verify**	Step 1: Send 6-digit email OTP for eligible Student ID	Student ID: 22-10301	1. Open VerifyAccountActivity 2. Enter ID 3. Tap Send Code	Calls /api/verify.php (send_code); transitions to Step 2	Masked school email & student preview shown; 60s timer starts	Pass	D-02
TC-M06	**MOD-03 Verify**	Step 2: Verify 6-digit code & test resend cooldown	6-digit code from email	1. Enter 6-digit code 2. Tap Verify Email	Calls /api/verify.php (verify_code); transitions to Step 3	Step 3 verified record summary displayed	Pass	D-02
TC-M07	**MOD-03 Verify**	Step 3: Set password & complete account setup	Password >=8 chars (letter + digit)	1. Enter & confirm password 2. Tap Create Account	Creates approved/verified account; shows Step 4 completion	Redirects to LoginActivity with Student ID prefilled	Pass	D-02
TC-M08	**MOD-04 Home**	Load greeting, plan card, 10 subject tiles & recent studies	Authenticated student	1. View HomeFragment 2. Pull to refresh	Populates stats, horizontal categories & recent research cards	All 10 categories & recent cards bound with 0 errors	Pass	D-04
TC-M09	**MOD-05 Search**	Search archive & filter by dynamic Subject Area & Year	Keyword + Category Spinner + Year Spinner	1. Open Research tab 2. Select subject & year	Fetches /api/research-list.php; updates results & count badge	Filtered cards & pagination controls updated	Pass	D-04
TC-M10	**MOD-06 Details**	Read full abstract on Free Access & view lock box	Free tier account, tap any research card	1. Open ResearchDetailActivity 2. Scroll abstract & access box	Full abstract & keywords visible; manuscript shows lock box	Lock box displays 'View Access Plans ->' button	Pass	D-03
TC-M11	**MOD-07 Plans**	Browse live access plans & current subscription status	Open Access Plans tab	1. Tap Access Plans in bottom nav	Loads Free (PHP 0), Research Access (PHP 99), Plus (PHP 249) from API	Plan cards, features & demo notice rendered	Pass	D-05
TC-M12	**MOD-07 Plans**	Activate paid plan via CheckoutActivity (Demo)	Select Research Access PHP 99/30d	1. Tap Get this plan 2. Check demo box 3. Tap Activate	Calls /api/checkout.php; shows receipt with SV- reference code	SessionManager updated; full access unlocked immediately	Pass	D-05
TC-M13	**MOD-08 Viewer**	Read protected PDF in ManuscriptViewerActivity	Active subscriber, tap Open Protected Reader	1. Open viewer 2. Test Prev/Next page & Zoom (75%–200%)	Streams PDF to internal cache; renders watermarked pages	Pages stamped with watermark; cache deleted on exit	Pass	D-03
TC-M14	**MOD-08 Viewer**	Enforce FLAG_SECURE & block unsubscribed access	Attempt screenshot or open viewer without plan	1. Attempt OS screenshot 2. Test expired subscription	OS blocks screenshot (FLAG_SECURE); server returns 403 if unpaid	Screenshot blocked; upgrade prompt shown for unpaid user	Pass	D-03

Test ID	Module	Function / Action	Precondition / Test Data	Test Steps	Expected Result	Actual Result / Evidence	Status	Issue ID
TC-M15	**MOD-09 Profile**	View verified student profile, plan & subscription history	Open Profile tab	1. Tap Profile in bottom nav 2. Pull to refresh	Displays full name, ID, email, BSIT program, status & history	All profile fields & SubscriptionHistoryAdapter populated	Pass	—
TC-M16	**MOD-09 Profile**	Sign out with confirmation dialog & token revocation	Tap Sign Out button	1. Tap Sign Out 2. Confirm dialog	Calls /api/logout.php to revoke mobile token; clears prefs	Token revoked on server; routed to LoginActivity	Pass	—

Status options: Pass / Fail / Blocked / Not Tested / N/A. Evidence may be a screenshot filename, screen-recording reference, test log, or concise observation.

D. ALPHA TESTING CHECKLIST — INTERNAL VERIFICATION

Module	Function	Valid Input	Invalid / Empty Input	Access / Permission Check	Error Handling	Result / Notes
Login	Student ID & Password Sign-In	Pass (24-01048)	Pass (Empty/Wrong rejected)	Pass (Pending/Rejected blocked)	Pass (Inline error banner)	Verified on Web & Mobile API
Verification	3-Step Student ID Email OTP	Pass (Valid BSIT ID + 6-digit OTP)	Pass (Invalid ID / expired OTP blocked)	Pass (Non-IT & already-linked blocked)	Pass (60s resend cooldown shown)	Verified via school_student_records
Home	Dashboard & Category Shortcuts	Pass (Loads stats, 10 categories, recent)	Pass (Empty search handled)	Pass (Requires valid student auth)	Pass (Retry button on network error)	All 10 subject tiles functional
Archive & Search	Search, Category/Year Filter & Sort	Pass (Keyword, category, year, sort)	Pass (No-match shows empty state)	Pass (Returns approved records only)	Pass (Clear filters resets state)	Pagination & suggestions verified
Research Details	Abstract & Access State Check	Pass (Full metadata & abstract shown)	Pass (Invalid ID shows error state)	Pass (Free=abstract; Paid=manuscript)	Pass (Lock reason clearly explained)	Related studies & keyword chips work
Plans & Checkout	Access Plans & Demo Activation	Pass (PHP 99/30d & PHP 249/90d activate)	Pass (Unchecked demo box blocked)	Pass (Authenticated student required)	Pass (Validation prompt displayed)	Generates SV-reference code & stacks days
Manuscript Viewer	Protected Watermarked PDF Reader	Pass (Renders pages + zoom/page nav)	Pass (Missing PDF handled gracefully)	Pass (HTTP 403 if no active plan)	Pass (FLAG_SECURE + watermark active)	No public URL or download button exposed
Profile & Logout	Account Info, History & Sign-Out	Pass (Shows identity, plan & history)	N/A	Pass (Revokes Bearer token & session)	Pass (Confirmation dialog before logout)	Session cleared cleanly

Use only checks relevant to each function. Examples: required fields, duplicate entries, boundary values, invalid formats, unauthorized access, confirmation prompts, and appropriate error messages.

E. BETA TESTING CHECKLIST — USER TASK VALIDATION

Tester ID	Module	Function / Task Attempted	Completed? (Y/N)	Ease (1–5)	Problem / Confusion	Suggested Improvement
_____	**Verification**	Verify Student ID via 6-digit school email code & set password	Y	5	None — 3-step progress indicator is clear	Keep 60s countdown visible on resend button
_____	**Login**	Sign in with verified Student ID and password	Y	5	None	Show/hide password toggle is helpful
_____	**Home & Discovery**	Browse IT subject areas and open a recent research study	Y	5	None	Subject record counts help identify active topics
_____	**Research Archive**	Search by keyword and filter by Subject Area and Year	Y	5	None	Autocomplete suggestions speed up searching
_____	**Research Details**	Read research abstract, keywords, and related studies on Free Access	Y	5	None — lock box clearly explains Free vs. Paid access	Clickable keyword chips are convenient

Tester ID	Module	Function / Task Attempted	Completed? (Y/N)	Ease (1–5)	Problem / Confusion	Suggested Improvement
_____	**Plans & Checkout**	Select Research Access (PHP 99 / 30 days) and complete Demo Checkout	Y	5	None — demo notice makes prototype nature clear	Reference code receipt confirms activation
_____	**Manuscript Viewer**	Read full watermarked PDF manuscript using zoom and page controls	Y	4	None	Zoom in/out and page counter work smoothly
_____	**Profile & Logout**	Review account details, subscription expiry, history, and sign out	Y	5	None	Logout confirmation prevents accidental sign-out

Suggested ease scale: 1 = Very difficult; 2 = Difficult; 3 = Acceptable; 4 = Easy; 5 = Very easy. Ask testers to describe what they attempted and what happened.

F. DEFECT AND RETEST LOG

Issue ID	Module / Function	Observed Problem	Severity	Steps to Reproduce	Evidence	Fix / Action	Retest Result
D-01	**Login / Session**	Mobile login sent 'student_number' instead of 'student_id' and lost session state	High	1. Log in from legacy mobile app 2. Call protected API	api/login.php & ApiClient.java	Aligned field to 'student_id'; added CookieJar + Bearer token auth	Pass (Fixed)
D-02	**Verification**	Legacy mobile app used deprecated open registration (HTTP 410) instead of 3-step OTP	High	1. Tap Register in legacy app 2. Submit name/password	api/verify.php & VerifyAccountActivity	Built 3-step Student ID email OTP verification API & native screen	Pass (Fixed)
D-03	**Manuscript Viewer**	Legacy mobile app exposed raw file_path & downloaded PDFs via DownloadManager	Critical	1. Open research detail 2. Tap Download PDF without plan	api/manuscript.php & ManuscriptViewerActivity	Gated PDF stream server-side; built native PdfRenderer viewer + FLAG_SECURE	Pass (Fixed)
D-04	**Home & Search**	Only 2 hardcoded categories appeared on mobile instead of 10 DB categories	Medium	1. Open Home/Search on mobile 2. Compare with web categories	api/home.php & SearchFragment.java	Loaded all 10 categories & years dynamically from MySQL	Pass (Fixed)
D-05	**Access Plans**	Mobile app had no Access Plans tab or demo checkout to unlock manuscripts	High	1. Try to upgrade plan from mobile app	PlansFragment & CheckoutActivity	Added 4th bottom-nav tab (Access Plans) & prototype CheckoutActivity	Pass (Fixed)
D-06	**Storage / Seed PDF**	Research record #1 PDF file was missing in storage/manuscripts/	Medium	1. Open Protected Reader for research_id=1	storage/manuscripts/	Provisioned research-20260918-07ae3eb4bf715cbb.pdf in storage/manuscripts/	Pass (Fixed)

Severity: Critical / High / Medium / Low. Record each issue once and reference its Issue ID in the function-level matrix.

G. MODULE COMPLETION SUMMARY

Module Name	Functions Listed	Tested	Passed	Failed	Blocked / N/A	Open Issues	Module Status
MOD-01 — Splash & Session Routing	4	4	4	0	0	0	Complete / Passed
MOD-02 — Native Login & Server Config	4	4	4	0	0	0	Complete / Passed
MOD-03 — 3-Step Student ID Verification	4	4	4	0	0	0	Complete / Passed
MOD-04 — Home Dashboard Fragment	4	4	4	0	0	0	Complete / Passed
MOD-05 — Research Archive & Search Fragment	4	4	4	0	0	0	Complete / Passed

Module Name	Functions Listed	Tested	Passed	Failed	Blocked / N/A	Open Issues	Module Status
MOD-06 — Research Details Activity	4	4	4	0	0	0	Complete / Passed
MOD-07 — Access Plans & Demo Checkout	4	4	4	0	0	0	Complete / Passed
MOD-08 — Protected Native PDF Viewer	4	4	4	0	0	0	Complete / Passed
MOD-09 — Profile Fragment & Logout	4	4	4	0	0	0	Complete / Passed

H. TEST CYCLE SUMMARY AND SIGN-OFF

Alpha result: [X] Proceed to Beta Testing [ ] Retest Required [ ] Major Rework Required

Beta result: [X] Acceptable for final review [ ] Acceptable after listed fixes [ ] Not acceptable yet

Outstanding issues / limitations: Prototype/Demo checkout mode (no live payment gateway connected); watermark deterrence.

Required actions and person responsible: Final demonstration & sign-off — _____

Role	Name	Signature	Date
Developer / Team Representative	_____	_____	_____
Testing Lead	_____	_____	_____
Instructor / Project Adviser	_____	_____	_____